
Digital Twins for Simulation and Security

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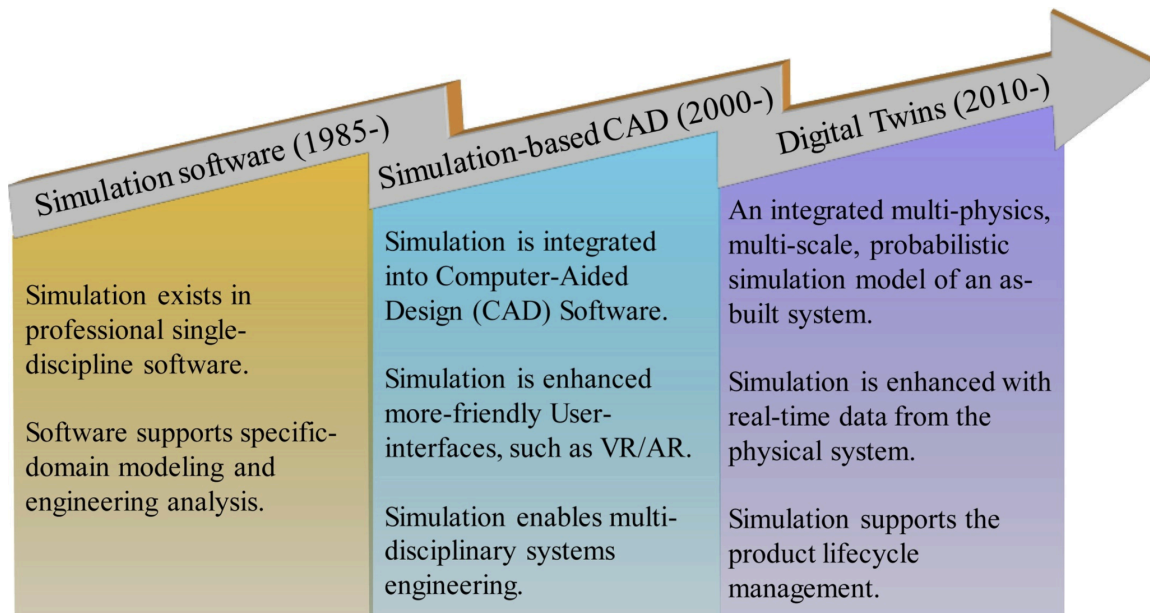
Introduction

This research set out to explore in-depth digital twins as a development in smart manufacturing, as well as its growing role in numerous other industries. It is then compared to other existing infrastructures in two similar security-related regards: Firstly, does digital twin modeling expand or contract the attack surface of a system that transfers to a digital twin system from more traditional decentralized operations? Secondly, can digital twins modeling for the primary sake of cybersecurity provide comparative value over other cutting edge security techniques? The answers to these questions will vary on a case by case basis, but across the board, digital twins can provide a relatively unique combination of pros and cons that present interesting value propositions in all kinds of industries. This is partially because of the inherent customizable nature of digital twins, which is more so a framework that can be adapted to meet extremely niche and sometimes totally unique needs of different systems.

Digital Twins

Digital twins have roots in the earliest NASA missions in the 1960s, in which they simulated Apollo 13 on Earth whilst it was on its way to and from the moon. The digital twins model as a commonplace in manufacturing, however, is a relatively recent development. This is likely due to the exponentially increasing speeds in advances made in computing, energy efficiency, and artificial intelligence. The idea has always remained to create a virtualized counterpart to a physical system or process, in which the properties of the original and the environment in which it exists are maintained as accurately as possible. In doing so, engineers can use these models to predict what might happen to a system if certain parameters are changed

without having to risk the time, money, and physical infrastructure that might have otherwise been required to carry out a more trial-and-error kind of approach. An aerospace engineer can test how an engine might react under certain risky climatic conditions without having to actually risk an expensive engine in a live test environment. Structural engineers can predict the necessary support a building might require in a given city and climate. Security engineers can duplicate an entire IT infrastructure, and run test attack scenarios to try and pinpoint weaknesses in a system rather than wait for a real attack (4). These are a few of many examples in which this ever-developing method of virtualization is continuing to shape the way companies research and develop new ways of doing things, or do the things they've always done in more optimized and lightweight ways.



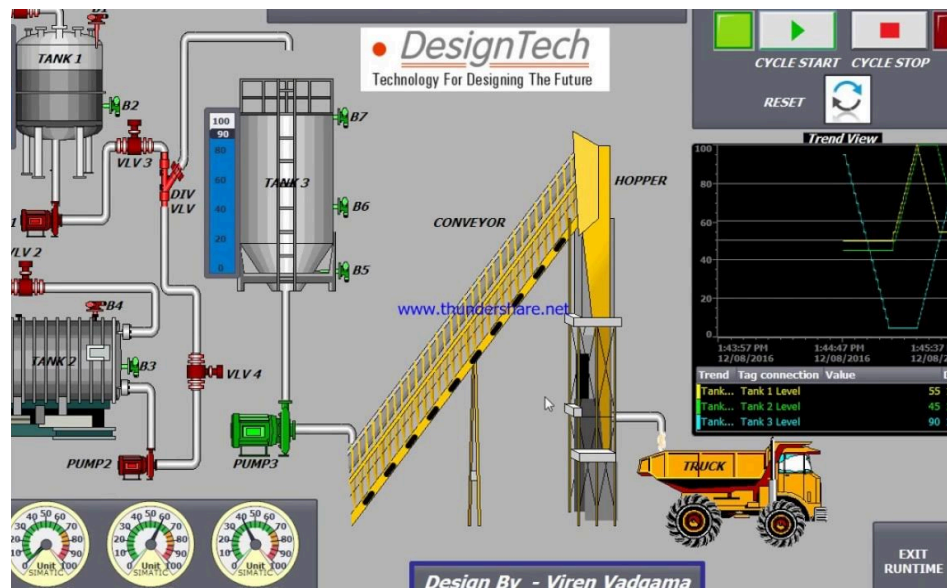
Evolution of Simulation Design Over Time (2)

Alternative Control Systems

To provide some context, there are many other commonly utilized methods to integrate and manage robotic systems: supervisory control and data acquisition (SCADA), internet of

things (IoT), simulation, and scaled down physical models to name a few. Each of these generally overlap with digital twins in one or many ways, but also have their differences.

SCADA is one of the closest relatives of digital twins, using real-time data to monitor and control systems while being able to monitor users if it notices unexpected behaviors. Generally, this can all be done remotely, which provides similar utility as digital twins, however SCADA systems tend to be much less complex, and are not able to accurately predict outcomes for new parameters (5). SCADA systems are metaphorically closer to a long list of if-then-else statements integrated with sensors and historical data, whereas digital twin models contain all that plus some implementation of how different parts of the system interact with each other and their environment. This allows predictive analysis for what might result for new input parameters.



Example SCADA Simulation (8)

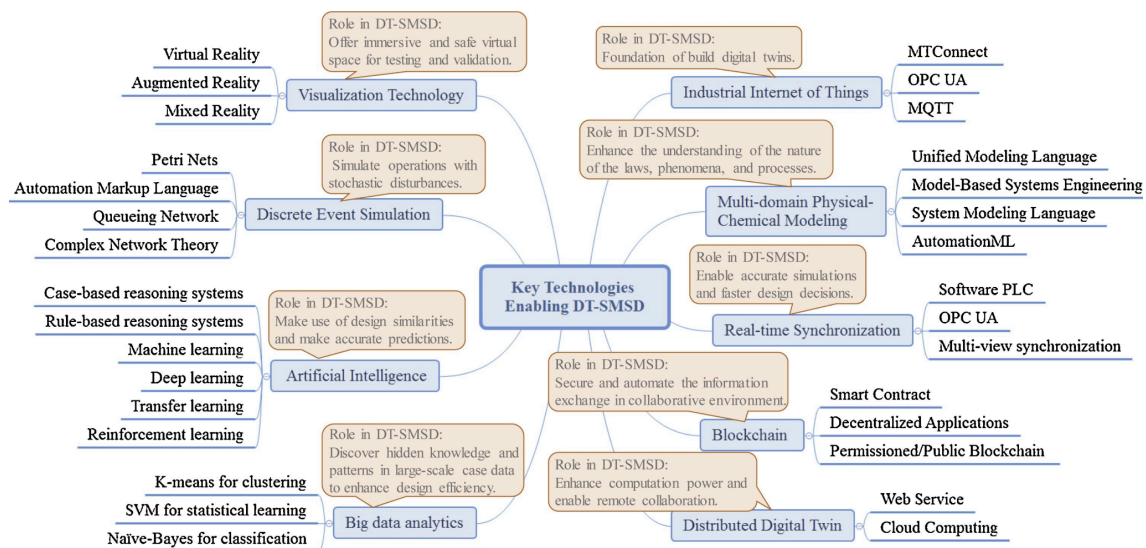
Similarly, IoT systems tend to focus on connecting large amounts of devices, collecting real time data, and automating as much as possible. The purpose of this is to efficiently allow for

remote monitoring and control over large networks and interconnected systems (6). It can be thought of as SCADA, but more so for digital devices connected to a larger network rather than physical machines with sensors connected to a larger system. There is not enough inherent complexity to simulate how a sophisticated cyberattack might affect a network, or how a system might tune its parameters to run in a more optimized way. IoT systems are great for what they are meant for, but for the higher upfront cost of establishing a digital twin, there is much greater potential to optimize along with it.

Simulation in some form is effectively a watered down digital twin. Most simulations that are not designed with the digital twin framework in mind are usually developed on a case by case, with the intention of maintaining or assessing particular parts of a system, with concern as to how change will affect the greater system as it relates to the particular problem it was developed for. Digital twins are meant to exist as a perpetual simulation of sorts, where every factor is accounted for in the twin and changes to parts can be tested, parts can be monitored and controlled. Cloud-computing has allowed for simulations to grow with the increased computing resources available to them in the cloud. If a simulation has all of the real time data and knowledge of the relationship between parts of the whole, and the computing power to predict how changes in parameters might affect the performance or outcome of a process, it is functionally just a digital twin itself.

Physical simulation has always been utilized to test systematic changes before their release, especially before computing was advanced enough to provide an effective alternative. Now that digital simulation in its various discussed forms is so widely used, however, it is usually a large waste of time and money to carry out physical tests when a simulation can achieve the same predictive analysis with a high accuracy. The argument for digital twins

remains the same: one gets more trials, more results, a better ability to predict, and more consistency. Being the most antiquated form of simulation in many ways, physical simulation is usually only used in life and death cases in which digital simulations are used in tandem (i.e. car and plane crash tests), as well as in marketing campaigns for the visual effect (a product supporting large weight), at which point the focus is no longer on creating an effective simulation but rather grabbing attention.



Digital Twins require integration of existing simulation tools (2)

Security Considerations of Digital Twins & Alternatives

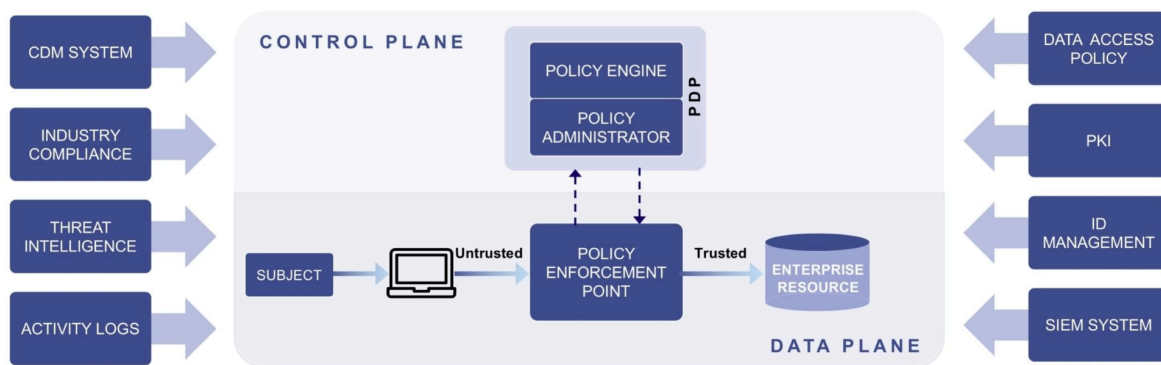
Now that an overview for digital twin technology has been established, and it has been placed next to a few other alternatives, it can be deduced that digital twin modeling is a culmination of different technologies to provide a real time synchronous event monitor that can integrate numerous components of a complex system such that it can also use data analytics and artificial intelligence to predict how a system might react under certain conditions. This is a very

centralized approach, and although there are vast benefits in allowing this interoperability between parts, a system can begin to approach a single point of failure if not careful.

For example, any system with a single point of failure runs the risk of bottlenecking if anything were to go wrong, but in practice, it is often safer to have everything in one system that is robustly secured, rather than multiple interconnected machines with different security standards. The centralized collection of real time data can allow for engineers to predict weaknesses in a system and run more accurately tuned simulations. The attack surface lengthens by this logic, as there becomes an increased reliance of the entire system on a central digital database on one or multiple machines. It is much more difficult to store backups as one would with a traditional database of login credentials or customer information, because of the demanding necessity of up to date feedback. A backup can be created of a digital twin, but without the feed from the system components, it is nothing more than an offline simulation. The idea of “rolling back servers” in this case might prove extremely difficult if the system contains hundreds of thousands of finely tuned physical machines rather than a handful of corrupt entries in a digital database.

Zero-trust architectures are commonly found in IoT systems. In these setups, no machines are programmed to trust one another until each exchange of information is initiated with a handshake and authentication. This doesn't provide complete security, but it mitigates the potential impact of malware on a system by making it difficult or impossible for malware to spread between machines. It also gives users time to try to isolate the malware and limit the damages. It is extremely difficult to implement an effective zero-trust architecture into a complex digital twin model without decreasing its effectiveness (3). There is a constant flow of information to and from multiple distinct integrated sources that often transcend any singular

type of structure or framework. If a centralized assembly line control system has an IoT system to integrate its physical machines and sensors, uses cloud computing to optimize its parameters, and historical data to make behavioral predictions all while collecting and storing real time information, it would be very computationally expensive to standardize everything such that every machine can safely communicate using a common zero-trust protocol (6). It is always a tradeoff between effectiveness and security of the digital twin model. This is true with just about any system configuration, but given that digital twins rely on complexity and a relatively huge amount of data transfer between machines, there is a threshold of complexity required for a digital twin to provide value enough to be worth the intense setup and difficulty to secure it.



Zero Trust Example Setup (7)

Comparatively, what SCADA systems lack in their ability to predict and simulate behaviors given real time and historical data, they have the potential to make up for in security. They are usually completely offline, and have very strict security protocols and firewalls, lessening the need to consider frameworks such as zero-trust, because there is not a regular exchange of information between the system and the outside world. In practice, this actually often leads systems to use outdated hardware and software longer than they should, given their

robust structure makes it difficult to implement updates regularly. This presents a new angle of security risk, as vulnerabilities in antiquated software and hardware are always piling up in publicly available websites online. An up to date SCADA system, however, is almost always easier to completely secure than a digital twin.

IoT systems tend to also be easier to secure than digital twins. The common theme of security versus capability remains true, as these systems tend to be more decentralized by nature, and oftentimes a breach in one machine doesn't mean easy access to the entire network. It is also common for IoT devices to use robust cryptographic schemes to allow for the secure exchange of information. Most IoT setups utilize zero-trust frameworks to varying degrees because that architecture often meets the needs of large networks of computers that are not constantly exchanging information, but need to be certain there are secure lines of communication for when they do. Although they are less trusting than digital twins or SCADA models, there is usually a significantly larger attack surface in IoT systems, and although lateral movement is more difficult for malware, there is still a much larger number of potential entry points that administrators have to be wary of, each with potentially their own operating systems, computing capabilities, and hardware configurations.

For comparison sake, physical real-world simulation is on the opposite side of the efficiency versus security spectrum relative to digital twins. There is nothing intrinsically digital about them, and if a company runs a test in an undisclosed location, the concern is just to keep any digital records secure. In that regard, the simulation itself has a nonexistent attack surface from a cybersecurity perspective, but is not scalable. Digital twins trade a lack of attack surface for intensely powerful scalability and repeatability. That being said, there is no evidence across the sources researched for this paper that suggests there exists an industry or cutting edge

problem in which physical simulation actively competes with digital simulation (let alone digital twins specifically). Instead, they are always used together to support one another. Oftentimes car companies will design a frame according to stress tests simulations, but won't sell the car until physical crash tests backup their results. It certainly stands to reason that the two opposite sides of efficiency versus security are often the choice combination to provide complete, reliable, repeatable, and secure results to an array of real world problems.

Security via Digital Twins

Digital twins are becoming increasingly powerful by enabling engineers to improve and optimize large complex systems. Instead of assessing the security of a digital twin model compared to others with the goal of improving the efficiency of a system, or improving a machine, digital twins can also be assessed as a cybersecurity tool themselves. They can be implemented with the purpose of monitoring a network, predicting weaknesses, and testing how a network might react to various types of cyberattacks. This concept takes the model that was developed for more industrial purposes and applies it to the newly and ever growing cybersecurity concerns.

Creating a digital replica of an existing network of machines allows teams to test updates before pushing them to production. There are countless examples of software not having been properly tested being pushed out to devices prematurely, causing any and all kinds of real world backups with effects that extend well beyond cyberspace. CrowdStrike is the most recent mainstream example of how consequential the lack of proper testing in a virtual environment can be. The ability to introduce new security protocol adjustments into a test environment is a crucial tool for developers. Smaller simulations can demonstrate how effective adjustments might be at a

more local level, but an accurately configured digital twin of an entire network is the only way to truly test scalability and performance under a set of fresh protocol adjustments without running the risk of pushing prematurely and having to deal with unforeseen consequences.

Like any application of digital twinning, users are given a replica with which they can monitor the original system. More importantly, they are given a sandbox duplicate of their process, product, or network with which reckless tests can be carried out and their effects observed. This is the epitome of penetration testing itself. A digital twin is the ideal penetration testing tool for engineers to be able to reasonably predict how to improve a system, to teach new engineers the intricacies of their network without having to protect the original, and to watch in real time if an exploit has only localized effects, or effects that spread throughout their network. They can simulate cyberattacks multiple times over to test different ways of dealing with malicious code, and get a better understanding of how to optimize their networks for security. Even if security has nothing to do with the goal of the network itself, the existence of its digital twin can multiply a team's ability to secure it.

Cybersecurity will always be a reactive industry, as new threats are always arising and the attacker is always one step ahead of the security engineers. Offensive penetration testers are extremely important for this reason, to try to catch vulnerabilities before they can be exploited. Digital twinning provides an entirely new surface for experimentation, where defensive and offensive security engineers alike can workshop ideas and watch how things react in real time, without risking the production environment.

Cyber Threats In Smart Manufacturing

As smart manufacturing takes up larger and growing quantities in industrial production, cyber threats will have to be addressed. Many threats are common to any infrastructure that exists primarily in the digital realm: data breaches, ransomware, and malware will gravitate towards any system that has important things depending on it. The most important threat to understand and mitigate in the case of digital twins are supply chain vulnerabilities. The focus cannot just be on protecting one large database and having backups. The nature of a production line that might employ a digital twin is that it is made up of large quantities of smaller machines that communicate. Assembly lines have always been made of up smaller parts, but the interconnectedness and deep integration of all of the parts associated with a setup employing digital twins means that even more robust security would be required to ensure that if one part of the supply chain is affected by malware, it has little/no ability to spread throughout the system. Keeping all of the software and firmware updated is the first way to secure a system, but having protocols in place to control access between parts of the whole is crucial in allowing the digital twin to receive information in real time while not being vulnerable to malicious code that might exist on a particular machine (1). In typical digital twin fashion, metrics can be monitored automatically and alerts can be raised if there is any unexpected behavior, not just from a performance perspective but also a security analysis perspective. If a machine is trying to communicate to a computer it isn't meant to interface with, or if data is being sent in an abnormal format or quantity, flags can be raised so that any potential threat can be analyzed manually.

Digital twinning will continue to grow and push its way into every sector of engineering that benefits from any form of simulation for testing. For now, the limiting factor remains how

difficult it is to configure an accurate twin, given the complexity of many systems that might benefit from one. As twinning improves and this becomes easier for more teams to utilize, digital twins will and already have proven to be extremely useful tools that take more effort to effectively secure, but provide immense value once established. As discussed, they will also take on a crucial role in cybersecurity, where their sandbox nature will allow for rapid testing and advancement in security for its underlying system. They can be described as catch-all or perfect simulations when executed correctly, and will continue to enable teams to research, develop, and execute in more secure environments in the future.

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